

A.3 随机数

标准库定义了一组随机数引擎类和适配器，使用不同数学方法生成伪随机数。标准库还定义了一组分布模板，根据不同的概率分布生成随机数。引擎和分布类型的名字都与它们的数学性质相对应。

这些类如何生成随机数的细节已经大大超出了本书的范围。在本节中，我们将列出这些引擎和分布类型，但读者需要查询其他资料来学习如何使用这些类型。

A.3.1 随机数分布

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除了总是生成 `bool` 类型的 `bernoulli_distribution` 外，其他分布类型都是模板。每个模板都接受单个类型参数，它指出了分布生成的结果类型。

分布类与我们已经用过的其他类模板不同，它们限制了我们可以为模板类型指定哪些类型。一些分布模板只能用来生成浮点数，而其他模板只能用来生成整数。

在下面的描述中，我们通过将类型说明为 `template_name<RealT>` 来指出分布生成浮点数。对这些模板，我们可以用 `float`、`double` 或 `long double` 代替 `RealT`。类似的，`IntT` 表示要求一个内置整型类型，但不包括 `bool` 类型或任何 `char` 类型。可以用来代替 `IntT` 的类型是 `short`、`int`、`long`、`long long`、`unsigned short`、`unsigned int`、`unsigned long` 或 `unsigned long long`。

分布模板定义了一个默认模板类型参数（参见 17.4.2 节，第 664 页）。整型分布的默认参数是 `int`，生成浮点数的模板的默认参数是 `double`。

每个分布的构造函数都有这种分布特定的参数。某些参数指出了分布的范围。这些范围与迭代器范围不同，都是包含的。

均匀分布

```
uniform_int_distribution<IntT> u(m, n);
uniform_real_distribution<RealT> u(x, y);
```

生成指定类型的，在给定包含范围内的值。`m`（或 `x`）是可以返回的最小值；`n`（或 `y`）是最大值。`m` 默认为 0；`n` 默认为类型 `IntT` 对象可以表示的最大值。`x` 默认为 0.0，`y` 默认为 1.0。

伯努利分布

```
bernoulli_distribution b(p);
```

以给定概率 `p` 生成 `true`；`p` 的默认值为 0.5。

```
binomial_distribution<IntT> b(t, p);
```

分布是按采样大小为整型值 `t`，概率为 `p` 生成的；`t` 的默认值为 1，`p` 的默认值为 0.5。

```
geometric_distribution<IntT> g(p);
```

每次试验成功的概率为 `p`；`p` 的默认值为 0.5。

```
negative_binomial_distribution<IntT> nb(k, p);
```

`k`（整型值）次试验成功的概率为 `p`；`k` 的默认值为 1，`p` 的默认值为 0.5。

泊松分布

```
poisson_distribution<IntT> p(x);
```

均值为 double 值 x 的分布。

884 **exponential_distribution<RealT> e(lam);**

指数分布，参数 lambda 通过浮点值 lam 给出；lam 的默认值为 1.0。

```
gamma_distribution<RealT> g(a, b);
```

alpha（形状参数）为 a ，beta（尺度参数）为 b ；两者的默认值均为 1.0。

```
weibull_distribution<RealT> w(a, b);
```

形状参数为 a ，尺度参数为 b 的分布；两者的默认值均为 1.0。

```
extreme_value_distribution<RealT> e(a, b);
```

a 的默认值为 0.0， b 的默认值为 1.0。

正态分布

```
normal_distribution<RealT> n(m, s);
```

均值为 m ，标准差为 s ； m 的默认值为 0.0， s 的默认值为 1.0。

```
lognormal_distribution<RealT> ln(m, s);
```

均值为 m ，标准差为 s ； m 的默认值为 0.0， s 的默认值为 1.0。

```
chi_squared_distribution<RealT> c(x);
```

自由度为 x ；默认值为 1.0。

```
cauchy_distribution<RealT> c(a, b);
```

位置参数 a 和尺度参数 b 的默认值分别为 0.0 和 1.0。

```
fisher_f_distribution<RealT> f(m, n);
```

自由度为 m 和 n ；默认值均为 1。

```
student_t_distribution<RealT> s(n);
```

自由度为 n ； n 的默认值均为 1。

抽样分布

```
discrete_distribution<IntT> d(i, j);
```

```
discrete_distribution<IntT> d{il};
```

i 和 j 是一个权重序列的输入迭代器， il 是一个权重的花括号列表。权重必须能转换为 double。

```
piecewise_constant_distribution<RealT> pc(b, e, w);
```

b 、 e 和 w 是输入迭代器。

```
piecewise_linear_distribution<RealT> pl(b, e, w);
```

b 、 e 和 w 是输入迭代器。

A.3.2 随机数引擎

标准库定义了三个类，实现了不同的算法来生成随机数。标准库还定义了三个适配器，可以修改给定引擎生成的序列。引擎和引擎适配器类都是模板。与分布的参数不同，这些引擎的参数更为复杂，且需深入了解特定引擎使用的数学知识。我们在这里列出所有引擎，◀ 885以便读者对它们有所了解，但介绍如何生成这些类型超出了本书的范围。

标准库还定义了几个从引擎和适配器类型构造的类型。`default_random_engine` 类型是一个参数化的引擎类型的类型别名，参数化所用的变量的目的是在通常情况下获得好的性能。标准库还定义了几个类，它们都是一个引擎或适配器的完全特例化版本。标准库定义的引擎和特例化版本如下：

default_random_engine

某个其他引擎类型的类型别名，目的是用于大多数情况。

linear_congruential_engine

`minstd_rand0` 的乘数为 16807，模为 2147483647，增量为 0。

`minstd_rand` 的乘数为 48271，模为 2147483647，增量为 0。

mersenne_twister_engine

`mt19937` 为 32 位无符号梅森旋转生成器。

`mt19937_64` 为 64 位无符号梅森旋转生成器。

subtract_with_carry_engine

`ranlux24_base` 为 32 位无符号借位减法生成器。

`ranlux48_base` 为 64 位无符号借位减法生成器。

discard_block_engine

引擎适配器，将其底层引擎的结果丢弃。用要使用的底层引擎、块大小和旧块大小来参数化。

`ranlux24` 使用 `ranlux24_base` 引擎，块大小为 223，旧块大小为 23。

`ranlux48` 使用 `ranlux48_base` 引擎，块大小为 389，旧块大小为 11。

independent_bits_engine

引擎适配器，生成指定位数的随机数。用要使用的底层引擎、结果的位数以及保存生成的二进制位的无符号整型类型来参数化。指定的位数必须小于指定的无符号类型所能保存的位数。

shuffle_order_engine

引擎适配器，返回的就是底层引擎生成的数，但返回的顺序不同。用要使用的底层引擎和要混洗的元素数目来参数化。

`knuth_b` 使用 `minstd_rand0` 和表大小 256。

索引

粗体页码指的是第一次定义该术语的页码, 斜体页码指的是各章“术语表”定义该术语的页码。

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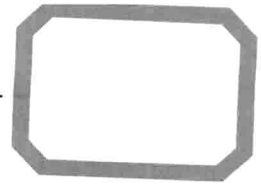
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